"Low-Cost IoT Retrofit for Urban Waste Monitoring and Route Optimization"

Team : InfraSense Systems

Theme : Environmental Sustainability

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Motivation

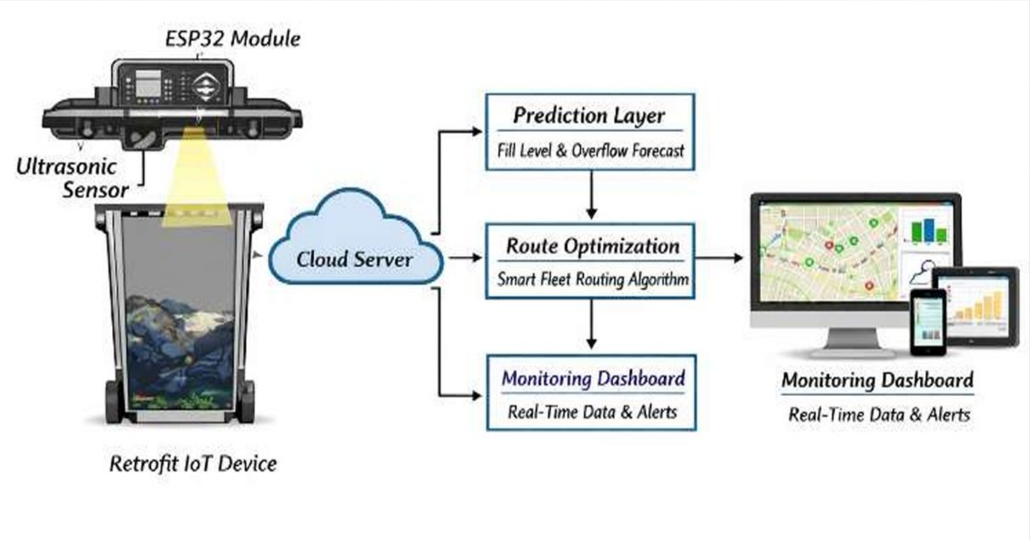
- Fixed schedules cause **overflowing bins, wasted fuel** and **pollution.**
- **Replacing** all bins with smart bins is **economically impractical**.

Problem Statement

Urban waste collection **lacks real-time intelligence**, causing **overflow** and **excess emissions**.

Novelty

- **Retrofits existing bins** — no replacement needed.
- **Predicts overflow** before it happens.
- Under ₹2,000 per unit.

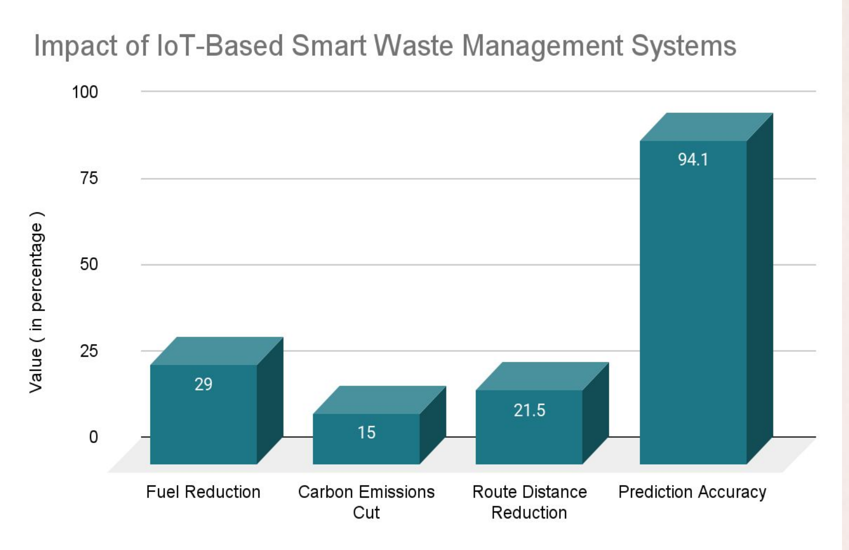


Methodology

- Hardware: **HC-SR04 Ultrasonic Sensor + ESP32** via Wi-Fi
- Fill% = (**(Bin Height – Distance) / Bin Height) × 100**
- Time Remaining = **(100 – Fill%) / Fill Rate**
- Priority Score = **(0.5 × Fill%) + (0.3 × Urgency)+(0.2×Distance)**

Market Potential

- 4,000+ urban local bodies in India
- **Swachh Bharat** & **Smart Cities** Mission funding available
- Sub ₹2,000 per unit — **scalable deployment**



Impact from Literature

- **29% fuel reduction** — Shafi et al. (2024)
- **21.5% average distance reduction** — Maciel et al. (2025)
- **94.1% overflow prediction accuracy** — AI-IoT-Graph Framework (2025)
- **15% carbon emissions cut** — Shafi et al. (2024)

Conclusion

We propose a **low-cost IoT retrofit** that makes existing bins smart. **Real-time** monitoring, **overflow prediction** and Dijkstra-based **route optimization** reduce fuel use, pollution and costs — **without replacing** any infrastructure.

References

1. **Addas et al. (2024)** — IoT Smart City Waste Management, *PLOS ONE*
2. **Maciel et al. (2025)** — IoT Routing Optimization Meta-Analysis, *Logistics, MDPI*
3. **Ghahramani et al. (2022)** — IoT Route Recommendation for Waste Management, *IEEE IoT Journal*
4. **Ahmed et al. (2023)** — IoT Intelligent Waste Management System, *Neural Computing and Applications*
5. **Frontiers (2025)** — AI-IoT Smart Waste Framework, *Frontiers in Sustainability*